Sample Document: Artificial Intelligence

1. Introduction

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks requiring human intelligence, such as understanding language, recognizing images, and solving problems.

2. Machine Learning

Machine Learning (ML) is a subset of AI. A machine-learning model learns patterns from examples in data. Common applications include spam detection, product recommendations, and forecasting.

3. Retrieval-Augmented Generation (RAG)

RAG combines information retrieval with a language model. First, the system searches documents for passages relevant to a question. Then it provides those passages to the language model as context to generate an answer grounded in the retrieved information.

4. Embeddings and Vector Stores

An embedding represents text as a list of numbers. A vector store saves embeddings and associated text so that semantically relevant passages can be retrieved.

5. Limitations

A RAG system can retrieve irrelevant information and generate incorrect answers. Good chunking, retrieval settings, and clear prompts can improve reliability. If the document does not contain an answer, the system should say so rather than inventing facts.